User Stories

Prompt

Act as a Business Analyst. Convert the Inventory Management System (IMS) requirements into user stories with acceptance criteria in Gherkin format (Given, When, Then). Ensure stories are testable, cover key business scenarios, and include relevant edge cases.

Response

Inventory Management System – User Stories with Gherkin Acceptance Criteria

Epic 1 – Authentication & User Management

User Story 1.1 – User Login

Story

As a user,
I want to log into the system securely,
So that I can access inventory features based on my role.

Acceptance Criteria

Feature: User Login

Scenario: Successful login with valid credentials
  Given the user is on the login page
  When the user enters a valid email and password
  And clicks the login button
  Then the system should authenticate the user

And redirect the user to the dashboard

Scenario: Login failure with invalid credentials
Given the user is on the login page
When the user enters an invalid email or password
And clicks the login button
Then the system should display an authentication error message

---

User Story 1.2 – Role-Based Access Control

Story

As an administrator,
I want users to access only authorized modules,
So that sensitive functionality is protected.

Acceptance Criteria

Feature: Role-Based Access Control

Scenario: Restricted module access
Given a warehouse staff user is logged in
When the user attempts to access the admin settings page
Then the system should deny access
And display an unauthorized access message

Epic 2 – Product Management

User Story 2.1 – Create Product

Story

As an inventory manager,
I want to create products,
So that inventory items can be tracked in the system.

Acceptance Criteria

Feature: Create Product

Scenario: Successfully create a product
  Given the inventory manager is on the product creation page
  When the manager enters valid product details
  And submits the form
  Then the system should create the product
  And assign a unique SKU

Scenario: Prevent duplicate SKU creation
  Given a product with the same SKU already exists
  When the manager submits the product form
  Then the system should reject the request
  And display a duplicate SKU error

User Story 2.2 – Update Product

Story

As an inventory manager,
I want to edit product information,
So that product data remains accurate.

Acceptance Criteria

Feature: Update Product

Scenario: Successfully update a product
  Given an existing product is available
  When the manager updates the product information
  And saves the changes
  Then the system should update the product details

User Story 2.3 – Search Products

Story

As a warehouse staff member,
I want to search for products,
So that I can quickly locate inventory items.

Acceptance Criteria

Feature: Search Products

Scenario: Search product by name
  Given products exist in the system
  When the user searches using a product name
  Then matching products should be displayed

Scenario: Search product by SKU
  Given products exist in the system
  When the user searches using a SKU
  Then the exact matching product should be displayed

Epic 3 – Inventory Management

User Story 3.1 – View Inventory Levels

Story

As an inventory manager,
I want to view current stock levels,
So that I can monitor inventory availability.

Acceptance Criteria

Feature: View Inventory Levels

Scenario: Display current inventory quantity
  Given products exist in inventory
  When the manager opens the inventory dashboard
  Then the system should display current stock quantities
  And warehouse locations

User Story 3.2 – Add Inventory Stock

Story

As warehouse staff,
I want to add stock quantities,
So that incoming inventory is recorded accurately.

Acceptance Criteria

Feature: Add Inventory Stock

Scenario: Add stock successfully
  Given a product exists in inventory
  When the user records a stock addition
  Then the system should increase the stock quantity
  And log the inventory transaction

User Story 3.3 – Remove Inventory Stock

Story

As warehouse staff,
I want to remove stock quantities,
So that dispatched inventory is reflected correctly.

Acceptance Criteria

Feature: Remove Inventory Stock

Scenario: Remove stock successfully
  Given sufficient stock exists
  When the user records a stock removal
  Then the system should reduce the inventory quantity
  And create a stock movement record

Scenario: Prevent negative inventory
  Given insufficient stock exists
  When the user attempts to remove stock
  Then the system should reject the transaction
  And display an insufficient stock message

User Story 3.4 – Transfer Inventory Between Warehouses

Story

As an inventory manager,
I want to transfer inventory between warehouses,
So that stock can be distributed effectively.

Acceptance Criteria

Feature: Warehouse Stock Transfer

Scenario: Transfer inventory successfully
  Given stock exists in the source warehouse
  When the manager transfers inventory to another warehouse
  Then the system should reduce stock from the source warehouse
  And increase stock in the destination warehouse
  And log the transfer transaction

Epic 4 – Supplier & Purchase Management

User Story 4.1 – Create Supplier

Story

As a procurement officer,
I want to create supplier records,
So that supplier information can be managed centrally.

Acceptance Criteria

Feature: Create Supplier

Scenario: Create supplier successfully
  Given the procurement officer is on the supplier page
  When valid supplier information is entered
  Then the system should save the supplier record

User Story 4.2 – Create Purchase Order

Story

As a procurement officer,
I want to create purchase orders,
So that inventory can be replenished.

Acceptance Criteria

Feature: Create Purchase Order

Scenario: Create purchase order successfully
  Given suppliers and products exist
  When the procurement officer creates a purchase order
  Then the system should generate a purchase order number
  And save the purchase order

User Story 4.3 – Receive Purchase Order

Story

As warehouse staff,
I want to receive purchased inventory,
So that stock levels are updated accurately.

Acceptance Criteria

Feature: Receive Purchase Order

Scenario: Receive inventory successfully
  Given an approved purchase order exists
  When the warehouse staff receives inventory
  Then the system should update inventory quantities
  And mark the purchase order as partially or fully received

Epic 5 – Stock Dispatch & Returns

User Story 5.1 – Dispatch Inventory

Story

As warehouse staff,
I want to dispatch inventory items,
So that outgoing inventory is tracked properly.

Acceptance Criteria

Feature: Dispatch Inventory

Scenario: Dispatch inventory successfully
  Given inventory stock is available
  When the warehouse staff processes a dispatch
  Then the system should reduce stock quantities
  And create a dispatch transaction

User Story 5.2 – Process Inventory Returns

Story

As warehouse staff,
I want to process returned items,
So that returned stock is added back into inventory.

Acceptance Criteria

Feature: Inventory Returns

Scenario: Process return successfully
  Given a previous dispatch transaction exists
  When returned inventory is processed
  Then the system should increase stock quantity
  And record the return transaction

Epic 6 – Reporting & Dashboard

User Story 6.1 – Generate Inventory Report

Story

As an inventory manager,
I want to generate inventory reports,
So that I can analyze stock data.

Acceptance Criteria

Feature: Generate Inventory Report

Scenario: Generate stock summary report
  Given inventory data exists
  When the manager requests a stock summary report
  Then the system should generate the report
  And allow export to PDF or Excel

User Story 6.2 – View Low Stock Alerts

Story

As an inventory manager,
I want to receive low stock alerts,
So that replenishment actions can be taken promptly.

Acceptance Criteria

Feature: Low Stock Alerts

Scenario: Trigger low stock notification
  Given a product reaches below reorder threshold
  When inventory levels are updated
  Then the system should generate a low stock alert

Epic 7 – Audit & Security

User Story 7.1 – Maintain Audit Logs

Story

As an administrator,
I want inventory changes to be logged,
So that inventory actions are traceable.

Acceptance Criteria

Feature: Audit Logging

Scenario: Log inventory updates
  Given a user updates inventory quantities
  When the transaction is completed
  Then the system should record:
    | Field |
    | User |
    | Timestamp |
    | Previous Value |
    | New Value |
    | Action Type |

User Story 7.2 – Session Timeout

Story

As a system administrator,
I want inactive sessions to expire automatically,
So that unauthorized access is minimized.

Acceptance Criteria

Feature: Session Timeout

Scenario: Expire inactive session
  Given a user is logged in

When the user remains inactive for the configured timeout duration
Then the system should automatically log out the user